Aalto University
School of Electrical
Engineering

Modeling Underwater Sound Propagation

Acoustics Seminar Project

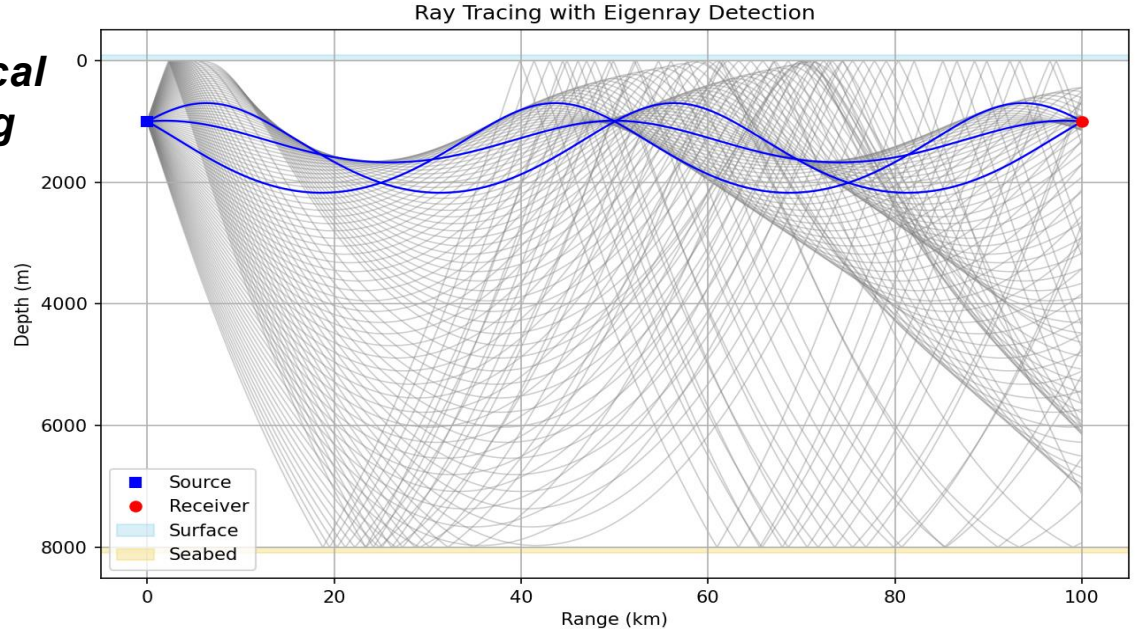
Hocine Montenez, Otto Ahoniemi & Atte Stenqvist

Aalto University
School of Electrical
Engineering

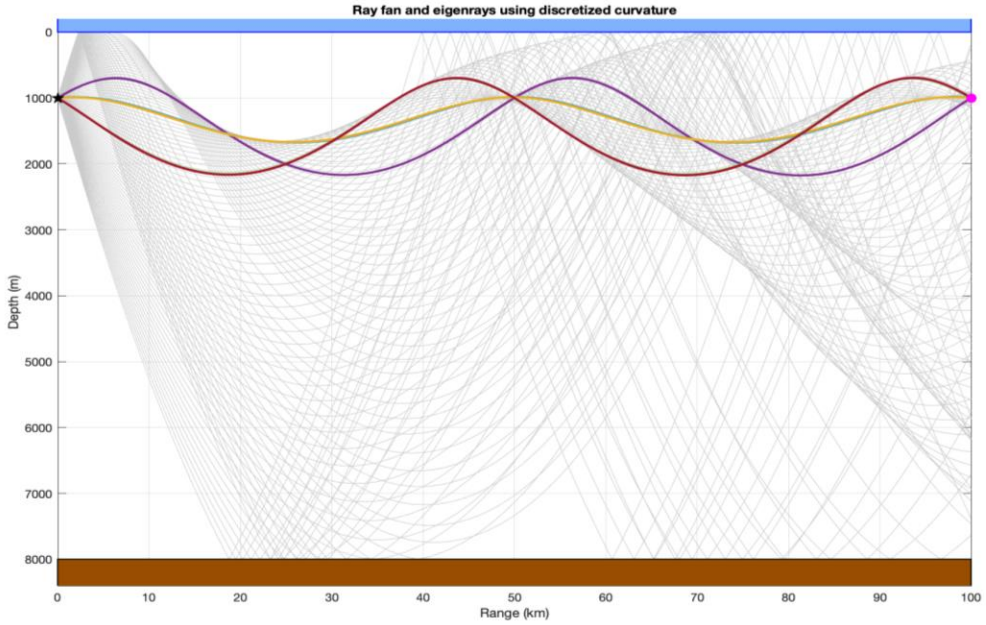
Where were we last time?

- Implementation of two different ray-tracing algorithms
- Bellhop model running
- Transmission loss for our models not yet implemented

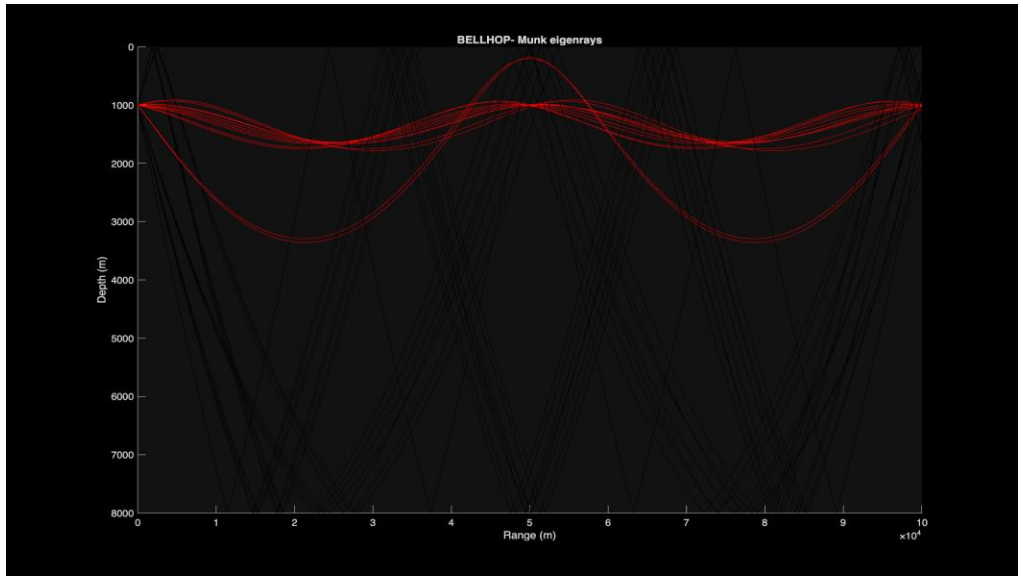
Geometrical ray tracing



c-linear



Bellhop



Transmission loss

- Describes difference in SPL between source and receiver
- Due to **spreading loss, absorption loss, reflection loss**, volume loss, surface scattering, bottom scattering, volume scattering
- We implemented the first three:

$$TL = TL_{spr} + TL_{abs} + TL_{\mathrm{ref}}$$

Transmission loss

Spreading loss: attenuation due to signal spreading in a lossless medium, spherical in near-field, cylindrical in far-field due to sea surface and seafloor acting as waveguide

$$TL_{spherical} = 20\log r,\ (r < z) \qquad TL_{\text{cylindrical}} = 10\log r,\ (r \gg z)$$

Absorption loss: how much acoustic energy is absorbed by the medium and turned into heat. Frequency-dependent, low frequencies are very mildly attenuated

$$A = A_0 \exp(-\alpha x) \qquad \alpha = \frac{1}{8.69}\left[3.3 \cdot 10^{-3} + \frac{0.11 f^2}{1 + f^2} + \frac{44 f^2}{4100 + f^2} + 3 \cdot 10^{-4} f^2\right]$$

Reflection loss: impedance mismatch determines how much is reflected and transmitted. Depends on seafloor material (sand, gravel, limestone, etc.) and incoming angle. Perfect reflection with water surface

$$R = \frac{Z_2 \cos\theta_i - Z_1 cos\theta_i}{Z_2 \cos\theta_i + Z_1 \cos\theta_\mathrm{i}}$$

Spreading loss variations

1. **Hybrid Spherical-Cylindrical Spreading (**implemented in c-linear model**)**

- Using spherical and cylindrical transmission loss assumption with transition range $r_t = z_{max}$

2. **Jacobian-based Geometric Spreading based on ray theory** (implemented in geometric ray tracing model)

- More accurate than simple 1/R models because it accounts for environmental refraction.
- Amplitude changes are determined by how sensitive a ray's position is to small changes in launch angle.

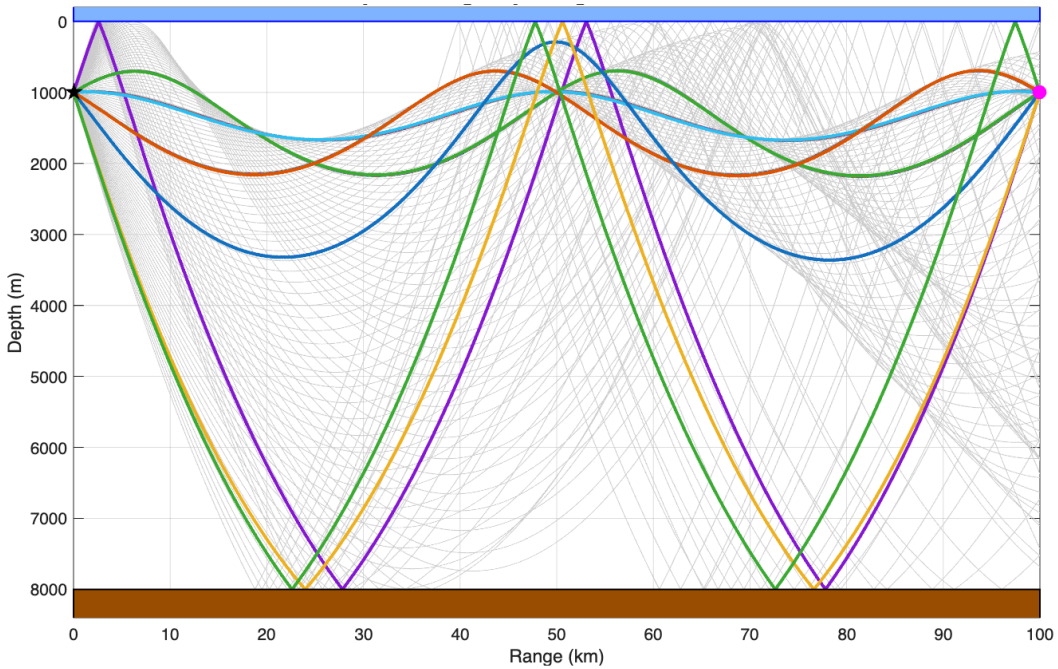
$$J(s) = \frac{r(s)}{\sin\theta\,(s)}\,\frac{\partial r(s)}{\partial\theta_0} \qquad A_0 = \frac{1}{4\pi}\left|\frac{c(s)\,\cos\theta_0}{c(0)\,J(s)}\right|^{1/2}$$

•If neighboring rays stay close = small J = focusing = larger amplitude
•If rays diverge = large J = defocusing = weaker amplitude

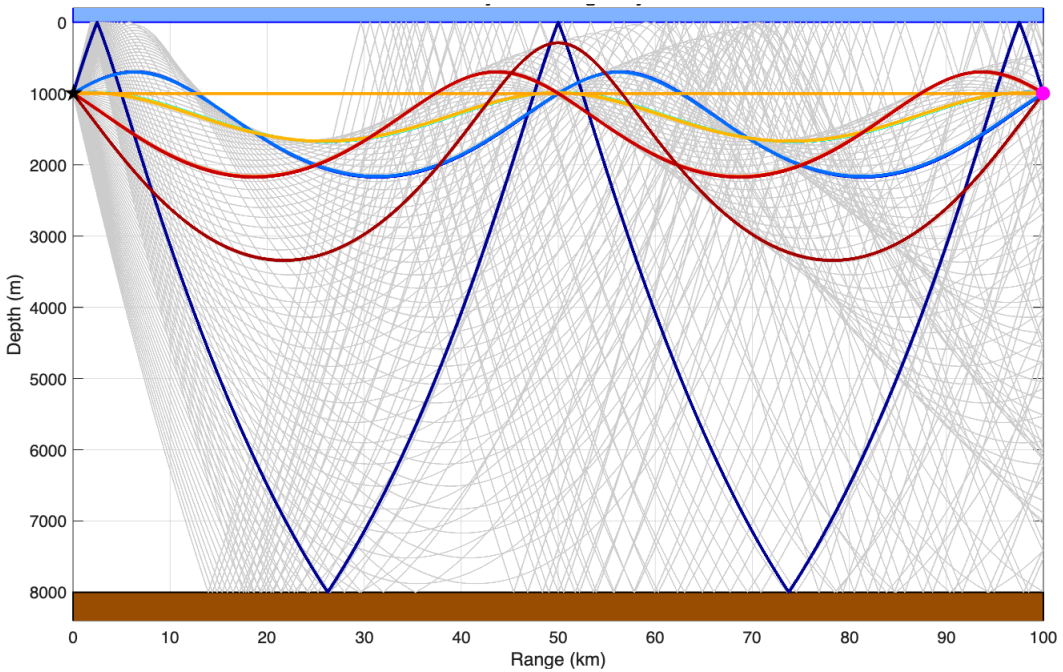
Aalto University
School of Electrical
Engineering

Comparing with Bellhop

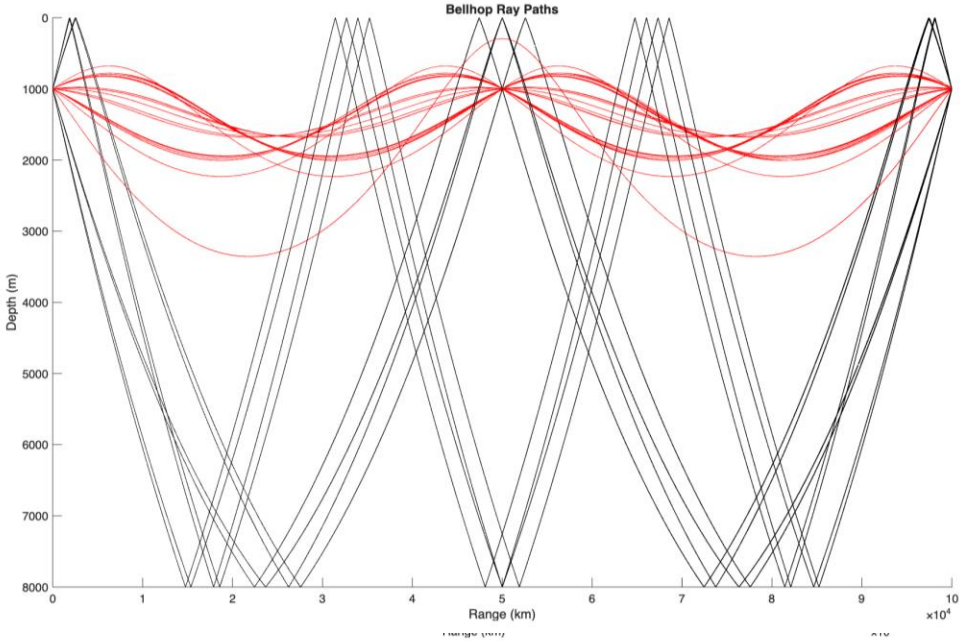
- Ray tracing, eigenray detection
- Impulse response at receiver



c-linear: 54 eigenrays



Geometrical ray tracing: 54 eigenrays

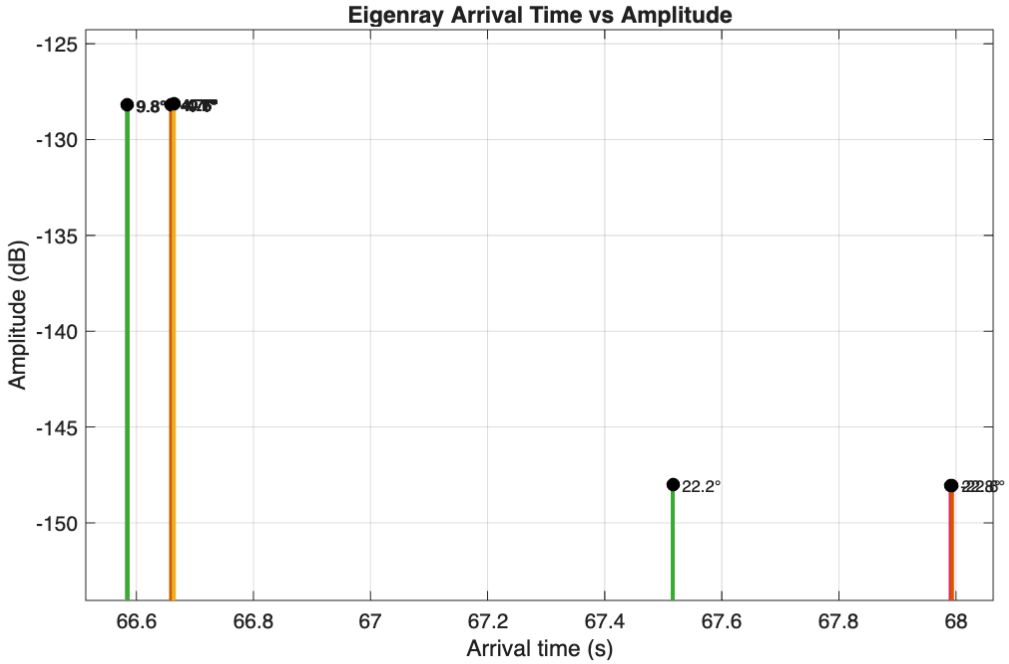


Bellhop: 24 eigenrays

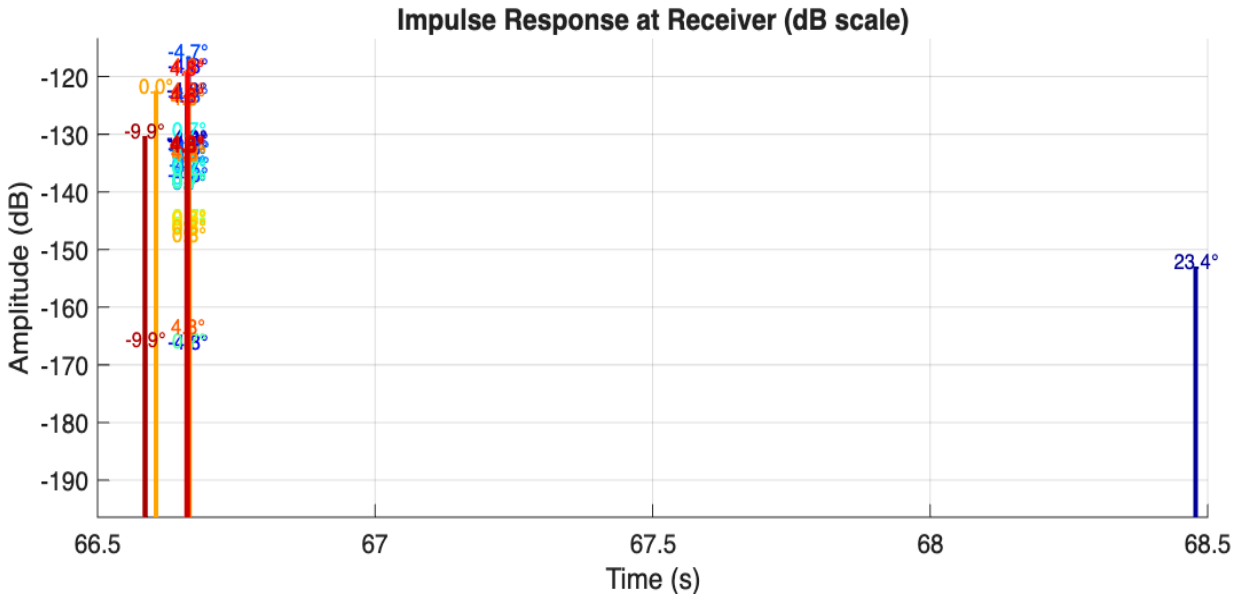
A!

Results: Impulse response at receiver

c-linear



Geometrical ray tracing



First ray to reach receiver is launched at around -9.8°, further path but faster speed of sound on edges of the sound channel axis!

Conclusions

- Eigenray count in custom models is high due to detection philosophy, dependent on detection tolerance
- Ray path characteristics are quite close to those in Bellhop
- Jacobian-based geometric spreading captures realistic spreading loss
- Frequency absorption is negligible at signal frequency of 100Hz

A!

Kiitos
aalto.fi